SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

QUESTIONS: 5 **Total Marks:50**

Annexure A
SENARIO BASED QUESTIONS

Criteria	Understanding of Scenario *(2)*	Identification of Key Issues *(2)*	Application of Concepts *(2.5)*	Decision Making *(2)*	Clarity of Response *(1.5)*	*Total(10)*
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I **A. MONESH / 192311430** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. MONESH

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A
Scenario based Questions

Q1. Data Preprocessing in Retail Analytics *(L3, L4)*

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	**Age**	**Gender**	**Monthly_Spend (₹K)**	**Purchase_Frequency**	**Loyalty_Level**
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

i. Identify missing values in **Age, Monthly_Spend, and Purchase_Frequency**
ii. Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
iii. Compute the imputed values and justify the selected method

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B. Outlier Detection and Treatment

i. Detect outliers in **Age and Monthly_Spend** using the **Z-score method**
ii. Identify records containing extreme values (e.g., Age = 105, Spend = 500)
iii. Apply suitable treatment methods (**capping / transformation / removal**) and justify

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C. Categorical Data Standardization

i. Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
ii. Convert all entries into a standardized format (**Male / Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics *(L4, L5)*

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
ii. Interpret whether the relationship is **positive or negative**
iii. Explain how covariance helps in identifying feature relationships

B. Feature Selection

i. Identify redundant attributes using **correlation or domain knowledge**
ii. Select an optimal subset of features for modeling
iii. Justify your selection

C. Data Transformation

i. Standardize the dataset (excluding **Cust_ID and Customer_Class**)
ii. Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics *(L3, L4, L5)*

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

i. Apply **3 bins** for Study_Hours and Test_Score
ii. Calculate bin width and define intervals

B. Equal-Frequency Binning

i. Divide data into **3 bins with equal records**
ii. Assign values accordingly

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High

- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:
200, 300,400, 600, 1000
(a) min-max normalization by setting min = 0 and max = 1
(b) z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis *(L3, L4, L5)*

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

📊 Sample Dataset: Visit Duration

Customer_ID	**Visit_Duration (seconds)**
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

✅ Tasks

A. Equal-Frequency Binning

i. Divide the dataset into **3 bins** with approximately equal number of records
ii. Assign each value to its respective bin
iii. Analyze how effectively this method groups short and long visit durations

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B. Z-Score Normalization

i. Compute the **mean and standard deviation** of visit duration
ii. Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**

C. Min-Max Normalization

i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

ii. Transform all values accordingly
iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

ANSWERS

Q1. Data Preprocessing in Retail Analytics

A. Handling Missing Values

i. Missing Values

Customer	Missing Attribute
C2	Age
C3	Monthly_Spend
C5	Purchase_Frequency

ii. Imputation Techniques

Method 1: Median Imputation

Age Values
23, 31, 105, 27, 40, 36, 42
Sorted:
23, 27, 31, 36, 40, 42, 105
Median = 36
So,
C2 Age = 36

Monthly Spend

20, 35, 500, 25, 45, 38, 300
Sorted
20,25,35,38,45,300,500
Median = 38
So,
C3 Monthly Spend = ₹38K

Purchase Frequency

5,8,10,50,12,9,45
Sorted
5,8,9,10,12,45,50
Median = 10
So,
C5 Purchase Frequency = 10

Method 2: k-NN Imputation

Nearest records are selected based on similar Age/Gender/Spend.
Example:
For C3, nearest customers are

- C6 (₹45K)
- C7 (₹38K)

Average
= (45+38)/2
= ₹41.5K
Similarly,
Age for C2 ≈ 35 years
Purchase Frequency for C5 ≈ 9

iii. Justification

- Median is robust against outliers.
- k-NN preserves similarity among customers and gives better estimates for recommendation systems.

B. Outlier Detection
i. Z-score Method
Formula

$$
Z=\frac{X-\mu}{\sigma}
$$

Age
Mean
= (23+36+31+105+27+40+36+42)/8
= 42.5
Standard deviation ≈ 24.2
For Age=105

$$
Z=\frac{105-42.5}{24.2}=2.58
$$

Since |Z| >2,
Age=105 is an outlier.

Monthly Spend
Mean
=(20+35+38+500+25+45+38+300)/8
=125.1
Standard deviation ≈169
Spend=500

$$
Z=\frac{500-125.1}{169}=2.22
$$

Outlier

Spend=300

$$
Z\approx1.03
$$

Not an extreme outlier.

ii. Extreme Records

Customer	Outlier
C4	Age=105
C4	Spend=500
C8	Spend=300 (high but acceptable)

iii. Treatment

Age 105

→ Cap at 80 years.

Spend 500

→ Winsorize (replace with upper percentile) or log transformation.

Reason:

Removes abnormal influence while retaining records.

C. Categorical Data Standardization

Original

M

Male

male

Female

F

FEMALE

Standardized

Original	Standard
M	Male
Male	Male
male	Male

Original	Standard
F	Female
Female	Female
FEMALE	Female

Q2. Covariance & Data Reduction
A. Covariance Analysis
Income
40,60,85,120,45,75,150,70
Purchase Value
100,150,210,320,110,180,360,160
Mean Income=80.625
Mean Purchase=198.75

Partial Calculation
For U1
[
(40-80.625)(100-198.75)
]
=(-40.625)(-98.75)
=4011.7
Similarly compute for all observations.
Total
≈ 22,863
Covariance
[
=\frac{22863}{7}
]
≈ 3266

ii. Interpretation
Covariance is positive.
As income increases,
purchase value also increases.

iii. Importance
Covariance
- identifies relationships

- detects redundant variables
- supports dimensionality reduction

B. Feature Selection
Highly correlated

- Income
- Purchase_Value
- Avg_Balance
- EMI

Choose

- Age
- Income
- Credit_Score
- Orders

Target
Customer_Class
Reason
These provide sufficient information with less redundancy.

C. Data Transformation
Standardization Formula

$$
Z=\frac{X-\mu}{\sigma}
$$

Apply to

- Age
- Income
- Purchase_Value
- Credit_Score
- Orders
- Avg_Balance
- EMI

Exclude

- Cust_ID
- Customer_Class

Why Standardization?

- Removes unit differences.
- Improves ML performance.
- Faster convergence.

- Important for KNN, PCA, SVM.

Q3. Data Transformation & Discretization
A. Equal Width Binning
Study Hours
Minimum=2
Maximum=11
Range=9
Bin Width
9/3=3Bins

Bin	Interval
B1	2–5
B2	5–8
B3	8–11

Assignments
B1
2,3,4,5
B2
6,7,8
B3
9,10,11

Test Score
Minimum=40
Maximum=100
Range=60
Bin Width
20
Bins
40–60
60–80
80–100
Assignments
40,50,55
65,75
85,90,95,98,100

B. Equal Frequency Binning
10 records

≈3 records per bin
Study Hours
Bin1
2,3,4
Bin2
5,6,7
Bin3
8,9,10,11

Test Scores
Bin1
40,50,55
Bin2
65,75,85
Bin3
90,95,98,100

C. Concept Hierarchy
Study Hours

Hours	Category
2–4	Low
5–7	Medium
8–11	High

Test Score

Score	Category
40–59	Poor
60–79	Average
80–100	Excellent

Q4. Normalization
Data
200,300,400,600,1000
Minimum=200
Maximum=1000

(a) Min-Max Normalization
Formula
[
X'=\frac{X-200}{800}
]

Value	Normalized
200	0.000
300	0.125
400	0.250
600	0.500
1000	1.000

(b) Z-score Normalization
Mean=500
Standard deviation≈282.84

Value	**Z-score**
200	-1.06
300	-0.71
400	-0.35
600	0.35
1000	1.77

Q5. Online Shopping Behavior Analysis
Visit Duration
15,30,45,60,120,300,600,900,1800,3600

A. Equal Frequency Binning
10 records
≈3 records/bin

Bin	Values
B1	15,30,45

Bin	Values
B2	60,120,300
B3	600,900,1800,3600

Analysis

- First bin contains short visits.
- Second bin contains medium visits.
- Third bin groups long visits, including extreme durations.

B. Z-score Normalization

Mean

$$\mu=\frac{7470}{10}=747$$

Standard deviation

≈1096

Formula

$$Z=\frac{X-747}{1096}$$

Duration	Z-score
15	-0.67
30	-0.65
45	-0.64
60	-0.63
120	-0.57
300	-0.41
600	-0.13
900	0.14
1800	0.96
3600	2.60

Discussion

- Extreme values (1800 and 3600 seconds) have high Z-scores.
- Z-score highlights outliers without restricting the data to a fixed range.

C. Min-Max Normalization

Formula

$$
X'=\frac{X-15}{3600-15}
$$

Duration	Min-Max Value
15	0.000
30	0.004
45	0.008
60	0.013
120	0.029
300	0.080
600	0.163
900	0.247
1800	0.498
3600	1.000

Comparison

Min-Max	**Z-score**
Values are scaled to the range 0–1	Values are centered around 0 with unit standard deviation
Sensitive to minimum and maximum values	Better for detecting outliers
Useful for neural networks and distance-based algorithms	Useful for statistical analysis and outlier detection